Johns Hopkins University (JHU) does not have a common core requirement for undergraduate students. There is a campus-wide writing requirement, but it can be satisfied by choosing from a wide variety of courses. There are no courses that all students in all majors must take; there are no books that all students must read. This is exacerbated in computer science (CS) where the freshman will start in different programming courses depending on their backgrounds, and students can customize their programs such that it would be rare for two students in the same class year to have taken the same exact set of courses. As a result, there is little in the academic curriculum to build community.

To alleviate this, in 2004 the Department of Computer Science established a freshman seminar course for its undergraduate majors only, 600.105 M&Ms: Freshman Experience, which is intended to build a cohort experience within the major, as well as provide an engaging introduction to the field of computing. This is a lightweight and informal 1-credit course, underscored by the passing of a candy tin around the room at the start of each class meeting. However it provides an opportunity to show students that computer science is about more than just programming. The course is offered in the fall semester only, and all freshman majors are expected to take it.

Grading for 600.105 is Pass/Fail only, based primarily on attendance. Faculty members in the department take turns presenting their research areas in two or three week blocks. In the past students have completed short homework assignments for each block as well, which then factored into their grade as a binary checkmark only. However, the assignments were not always particularly thought provoking and primarily existed to insure some accountability that students absorbed the material that had been presented. Accordingly, the course did not have a coherent take-away for the students.

In July 2013 the SIGCSE mailing list had a thread on books that could be used to introduce computing to a non-technical audience [3]. Although the main usage seemed to be for social implications of computing oriented general education courses, some had been used in pilot courses for the Advanced Placement Computer Science Principles course. Similarly, Leigh Ann Sudol's work on using reading assignments in an introductory programming course was focused on the connections students made between the computing concepts and their everyday lives [4]. For her assignments students could choose from lists of books to read, rather than all reading the same thing.

Based on the common core reading requirement across other campuses as well as these uses of computing related books in computer science education, in 2013 the department introduced a common book component to the M&Ms course for the CS freshmen with a three-fold goal: 1) enhance the cohort experience for CS majors, 2) replace several short assignments with meaningful homework that would become a lasting take-away from the course, and 3) use the book choice to introduce students to some aspect of the culture of computing.

IMPLEMENTATION

In keeping with the lightweight nature of the course and the varied backgrounds of these starting students, the book was chosen to be relatively short and not too technical, but clearly introducing some aspect of the culture of the field. Students were given most of the semester to read it. A short (2-3 page) opinion paper was due at the last class

which was then devoted to discussing the book. The faculty course coordinator a third year grad student relatively close to the students' ages prepared and led this session.

The course typically has 30-35 students in it each year, and each class session is 50 min. During the final class the students were broken into small groups of 4-5 based on where they were sitting in the classroom. The discussion leaders prepared a few questions to structure these small group discussions, and incrementally presented them among the student groups during this time, ``eavesdropping" to get hints as to how conversations were going, interject some new ideas, and make sure that the students stayed on task.

Approximately 30 minutes were spent in the small groups, leaving sufficient time for sharing with the entire class afterwards. Each group had the opportunity to present their joint answers to the discussion questions in a round-robin fashion, with the discussion leaders clarifying, summarizing and occasionally challenging the students throughout. After the small group discussions, the leaders asked the students eleven questions to assess their individual backgrounds and mindsets, based on points raised by the book. These were answered in a peer-blind manner (students had to close their eyes and raise their hands) to prevent peer pressure from influencing their answers.

In the inaugural year of this approach, and in light of its recent 10th anniversary, the faculty thought it fitting to tackle gender issues head-on by having the students read Jane Margolis and Allan Fisher's work Unlocking the Clubhouse: Women in Computing [2]. An additional goal in choosing this particular book was to gain insight as to what extent the issues revealed by the study were still applicable to their students today. They also wanted the male undergraduates to be aware of the issues that have persisted, in order to build a more welcoming community for women. In recent years the participation of undergraduate women in the CS department has increased to 23% of the majors and 44% of the minors across all class years. These numbers continue to rise based on higher ratios of women in the lower years, and the goal is to support that in as many ways as possible. For example, the class of 2016 is currently the largest population of majors, 31% of whom are women.

OUTCOMES

The department introduced the book assignment and discussion session into the freshman seminar course with three goals in mind. In addition, the inaugural book selection introduced a fourth objective, to essentially take the temperature of these freshmen regarding gender issues. While rigorous evaluation of these goals is not possible at this point in time, the results are encouraging enough to warrant further study.

Goal 1: Enhance The Cohort Experience

Prior offerings of the M&Ms course indicate that it does provide the freshmen majors with a cohort experience that had previously been lacking. In standard course evaluations, getting to know the other freshman majors is usually cited as one of the best aspects of the course. Upperclassmen who took the course have been overheard asking